BACHELOR OF COMPUTER APPLICATIONS

SEMESTER 1

FUNDAMENTALS OF MATHEMATICS

Unit 3

Limit and Continuity

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1. INTRODUCTION

In Unit 2, we embarked on an enlightening journey through the world of composite functions, a domain where the elegance of mathematics truly unfolds. Composite functions, at their core, represent the amalgamation of two distinct functions, creating a third function that encapsulates the sequential application of the first two. This concept mirrors the procedural steps in computer algorithms, where the output of one process seamlessly transitions into the input of another, showcasing the interconnectedness of mathematical operations.

But our exploration didn't stop there. We delved deeper into the arithmetic of functions, examining how operations like addition, subtraction, multiplication, and division play out when applied to functions. It's akin to mixing ingredients in precise measures to concoct a new recipe, each operation unveiling new flavours of mathematical functions.

Furthermore, we ventured into the rhythmic world of periodic functions, unveiling the repetitive patterns that underpin phenomena like the oscillations of a pendulum or the cycles of the moon. This segment highlighted the predictability and cyclicity inherent in certain functions, echoing the recurrent nature of events in both mathematics and the real world.

The concepts of even and odd functions added another layer to our understanding, illustrating how functions mirror themselves across axes or pivot around origins, much like symmetrical patterns in nature or architecture.

As we transition into Unit 3, the focus shifts towards the foundational concepts of limits and continuity—a pivotal area of study for any aspiring computer application specialist. Limits, in their abstract beauty, allow us to grasp the behaviour of functions as they approach a particular point, shedding light on the tendencies of functions under constrained conditions. This concept is crucial for understanding algorithms' efficiency and behaviours, especially when dealing with near-boundary conditions in computing tasks.

Continuity, on the other hand, ensures a seamless transition within functions, an attribute vital for ensuring that algorithms and software functions perform without unexpected disruptions. Understanding these concepts is akin to mastering the art of predicting and

ensuring smooth transitions in computational processes, a skill indispensable in the realm of computer applications.

Navigating through these complex mathematical concepts requires a strategic approach, blending theoretical understanding with practical application. It's essential to start with a solid grasp of basic definitions and properties, gradually layering on complexity through examples and exercises. Visual aids such as graphs and diagrams play a crucial role in demystifying abstract concepts, making them more accessible.

Collaboration and discussion with peers can illuminate different perspectives, enhancing comprehension. Moreover, applying these mathematical concepts to real-world scenarios and computational problems can bridge the gap between theory and practice, making the learning process both relevant and engaging.

1.1 Objectives

In this topic we will learn about,

- *Recall the fundamental definitions of limit and continuity in mathematics*
- *Explain the concept of limits as the behaviour of functions near a particular point*
- *Apply the first principle of limits (ε-δ definition) to evaluate limits of functions at specific points*
- *Analyse the continuity of functions by examining their behaviour around specific points*

2. LIMIT AND CONTINUITY

Definition of Limit

In mathematics, the limit of a function represents the value that the function approaches as the input gets closer and closer to a particular point. It's a fundamental concept for understanding the behaviour of functions, especially when dealing with values that might not be defined at that specific point.

The notion of a limit is a fundamental cornerstone in the field of calculus, acting as a crucial link between the realm of mathematical functions and the empirical events they represent. The concept of limit characterises the behaviour of a mathematical function when its input values converge towards a certain point. Mathematically speaking, the concept of a limit of a function at a certain point refers to the value that the function tends to approach as the input values progressively approach that particular point.

In a more formal context, we shall examine a function denoted as f(x) and a specific point referred to as 'a'. The limit of the function f(x) as x approaches a, represented as lim(x→a) f(x), is defined as the value L such that for any positive ε (epsilon), there exists a corresponding positive δ (delta) such that if the absolute difference between x and a (|x - a|) is smaller than δ, then the absolute difference between f(x) and L (|f(x) - L|) is smaller than ε. This statement encompasses the concept that by selecting a value of 'x' in close proximity to 'a', we have the ability to approach the desired value 'L' for the function f(x) to an arbitrary degree.

A function $y = f(x)$ is said to tend to limit l as x tends to a, if the difference between the values of $f(x)$ and l becomes smaller and smaller, whenever the difference between the values of x and a is smaller and smaller.

We write it symbolically as

$$\lim_{x \to a} f(x) = l$$

Definition of Continuity

A function is said to be continuous at a point 'a' if the limit of the function at 'a' exists and is equal to the value of the function at 'a'. In other words, there are no sudden jumps or breaks in the graph of a continuous function.

Continuity is a fundamental attribute of functions that guarantees a seamless and uninterrupted linkage between locations inside their respective domains. A function is deemed to exhibit continuity at a certain point if there are no sudden discontinuities or discontinuities in its graph at that moment. In more accessible language, it is possible to trace the graph of a continuous function without the need to raise one's writing instrument, therefore demonstrating a smooth and uninterrupted transition from one value to another.

In order for a function to exhibit continuity at a given point 'a', it is necessary for three specific requirements to be satisfied:

It is necessary for the function to have a well-defined value at the point 'a'.

The existence of the limit of the function as x approaches 'a' is a necessary condition.

It is necessary for the value of the function at 'a' to be equivalent to the value of the limit.

The concept of continuity guarantees that when the independent variable approaches a specific point, the function's values do not experience abrupt alterations, but rather exhibit a seamless transition from one value to the subsequent. The comprehension of this notion has significant importance in comprehending the behaviour of functions across many mathematical domains, encompassing calculus, analysis, and its applications in physics, engineering, and other related fields.

Example 1: Check the continuity of the function f given by f (x) = 2x + 3 at x = 1.

Solution: First note that the function is defined at the given point x = 1 and its value is 5. Then find the limit of the function at x = 1. Clearly

$$\lim_{x\to 1} f(x) = \lim_{x\to 1}(2x+1) = 2(1)+3 = 5$$

Thus

$$\lim_{x\to 1} f(x) = 5 = f(1)$$

Hence, f is continuous at x = 1.

Example 2: Examine whether the function f given by f (x) = x^2 is continuous at x = 0.

Solution: First note that the function is defined at the given point x = 0 and its value is 0. Then find the limit of the function at x = 0. Clearly

$$\lim_{x\to 0} f(x) = \lim_{x\to 0} x^2 = 0^2 = 0$$

Thus,

$$\lim_{x\to 0} f(x) = 0 = f(0)$$

Hence, f is continuous at x = 0.

Example 3: Discuss the continuity of the function f given by f(x) = | x | at x = 0.

Solution: By definition

$$f(x) = \begin{cases} -x, x < 0 \\ x, x \geq 0 \end{cases}$$

Clearly the function is defined at 0 and f (0) = 0. Left hand limit of f at 0 is

$$\lim_{x\to 0^-} f(x) = \lim_{x\to 0^-} (-x) = 0$$

Similarly, the right-hand limit of f at 0 is

$$\lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} x = 0$$

Thus, the left-hand limit, right hand limit and the value of the function coincide at

x = 0. Hence, f is continuous at x = 0.

3. FIRST PRINCIPLE OF LIMITS (E-Δ DEFINITION)

The first principle of limits, also known as the ε-δ definition, provides a formal way to express the concept of limit. It states that for a function f(x), the limit as x approaches a point 'a' is 'L' if, for any positive ε (epsilon), there exists a positive δ (delta) such that if the distance between x and a (|x - a|) is less than δ, then the distance between f(x) and L (|f(x) - L|) is less than ε.

The First Principle of Limits, also known as the - (epsilon-delta) definition, provides a logical and accurate approach to express the concept of limits in the field of calculus. This concept establishes the fundamental basis for comprehending the behaviour of a function when its input values converge towards a certain point. The mathematical framework presented in this study provides a means of quantifying the concept of "closeness" between input values and the corresponding function values.

Let us consider a function f(x) and a specific point 'a' at which we aim to compute the limit. According to the given definition, the limit of the function f(x) as x approaches an is L if, for each arbitrarily small positive value of ε (epsilon), there exists a matching positive value of δ (delta) such that the function values are also within a specified closeness to L. As the independent variable approaches a specific value 'a', the corresponding values of the function tend to converge towards a limit 'L' within a predetermined proximity.

Mathematically, the expression of this connection is as follows:
For every $\varepsilon > 0$, there exists a $\delta > 0$ such that if $0 < |x - a| < \delta$, then $|f(x) - L| < \varepsilon$.

For each given positive value of ε, there exists a positive value of δ such that if the absolute value of the difference between x and an is less than δ and more than zero, then the absolute value of the difference between f(x) and L is less than ε.

The crux of this formulation is in the comprehension that by judiciously choosing a suitable δ for any given ε, we may regulate the proximity of the function values to the limit 'L' as the input values approach 'a'. The ε-δ definition effectively encapsulates the concept of a limit by rigorously formalising the idea of "approaching arbitrarily close" through the use of mathematical inequalities. This concept forms the basis for the analysis of limits and plays a

crucial role in defining the fundamental principles of calculus and its applications across many domains of mathematics and science.

Illustration of concept of limit

Consider the function $\boldsymbol{y = f(x)}$ defined by $\boldsymbol{y = f(x) = 2x + 4}$

Table - 1		
$\boldsymbol{x}$	$\boldsymbol{y = 2x + 4}$	$\boldsymbol{y - 6}$
0.9	5.80	0.2
0.95	5.95	0.05
0.99	5.96	0.04
0.995	5.99	0.01
0.999	5.998	0.002
0.9999	5.9998	0.002
:	:	:
:	:	:

4. PROPERTIES OF LIMITS

Sum Rule: The limit of the sum of two functions is the sum of their individual limits.

The Sum Rule is a fundamental property that governs the behaviour of limits when dealing with the addition of functions. According to this rule, the limit of the sum of two functions, say f(x) and g(x), as x approaches a particular point 'a', is equal to the sum of their individual limits at that point. Mathematically, it can be expressed as:

$$\lim_{x \to a}[f(x) + g(x)] = \lim_{x \to a} f(x) + \lim_{x \to a} g(x)$$

This property underscores the fact that limits respect the arithmetic operation of addition, allowing us to evaluate the limit of a sum by evaluating the limits of the individual functions.

Product Rule: The limit of the product of two functions is the product of their individual limits.

The Product Rule is a counterpart to the Sum Rule, governing limits involving multiplication of functions. For two functions f(x) and g(x), the limit of their product as x approaches 'a' is the product of their individual limits at that point:

$$\lim_{x \to a}[f(x) * g(x)] = \lim_{x \to a} f(x) * \lim_{x \to a} g(x)$$

This rule highlights that the limit of a product can be found by evaluating the limits of the individual functions and then multiplying them together.

Difference Rule: The limit of the difference between two functions is the difference of their individual limits.

The Difference Rule deals with the limit of the difference between two functions. It states that the limit of the difference between two functions f(x) and g(x) as x approaches 'a' is the difference of their individual limits:

$$\lim_{x \to a}[f(x) - g(x)] = \lim_{x \to a} f(x) - \lim_{x \to a} g(x)$$

This rule emphasizes that limits follow the rules of subtraction when applied to functions.

Quotient Rule: The limit of the quotient of two functions is the quotient of their individual limits, provided the denominator's limit is not zero.

The Quotient Rule pertains to the limit of the quotient of two functions. Given functions f(x) and g(x), where the limit of g(x) as x approaches 'a' is not equal to zero, the limit of their quotient as x approaches 'a' is the quotient of their individual limits:

$$\lim_{x\to a}[f(x)\ /\ g(x)] = \left(\lim_{x\to a} f(x)\right) / \left(\lim_{x\to a} g(x)\right),\ if\ \lim_{x\to a} g(x) \neq 0$$

This rule is valid when the denominator's limit is nonzero, ensuring that division by zero is avoided.

These properties of limits play a crucial role in simplifying limit evaluations, making calculations more manageable and aiding in the analysis of complex functions by breaking them down into simpler components.

L'Hôpital's Rule

L'Hôpital's Rule is a powerful tool in calculus that helps evaluate the limits of indeterminate forms by using derivatives. Indeterminate forms are expressions that do not yield a clear limit value when directly substituted into a limit. The rule is particularly useful for limits involving fractions where both the numerator and the denominator approach zero or infinity. L'Hôpital's Rule states that if the limit of the ratio of the derivatives of the numerator and denominator exists as x approaches a specific point 'c', then the limit of the original fraction also exists and is equal to the value of the ratio of the derivatives. Mathematically, the rule can be stated as follows:

If $\lim_{x\to c}[f(x)\ /\ g(x)]$is an indeterminate form 0/0 or ∞/∞, and both,

$\lim_{x\to c}[f'(x)\ /\ g'(x)]$ and $\lim_{x\to c}[g'(x)]$

are finite (where f'(x) and g'(x) represent the derivatives of f(x) and g(x)), then:

$$\lim_{x\to c}[f(x)\ /\ g(x)] = \lim_{x\to c}[f'(x)\ /\ g'(x)]$$

L'Hôpital's Rule provides a systematic method for finding the limit of these indeterminate forms by simplifying the expression using derivatives, ultimately revealing the limit value.

Example:

Consider the limit:

$$\lim_{x \to 0} \left[\frac{sin(x)}{x} \right]$$

Direct substitution gives an indeterminate form 0/0. Applying L'Hôpital's Rule, we differentiate the numerator and denominator with respect to 'x': Numerator: Derivative of sin(x) is cos(x). Denominator: Derivative of x is 1.

So,

$$\lim_{x \to 0} \left[\frac{sin(x)}{x} \right] = \lim_{x \to 0} \frac{cos(x)}{1} = \frac{cos(0)}{1} = 1$$

Here, L'Hôpital's Rule is applied when we have an indeterminate form 0/0. By taking derivatives of the numerator and denominator, we transform the limit into a form that can be directly evaluated. In this example, as x approaches 0, the limit of sin(x) / x becomes 1.

Example:

Consider the limit:

$$\lim_{x \to \infty} \left[\frac{x}{e^x} \right]$$

Direct substitution gives an indeterminate form ∞/∞. Using L'Hôpital's Rule, we differentiate the numerator and denominator with respect to 'x': Numerator: Derivative of x is 1. Denominator: Derivative of e^x is e^x.

So,

$$\lim_{x \to \infty} \left[\frac{x}{e^x} \right] = \lim_{x \to \infty} \frac{1}{e^x} = 0$$

L'Hôpital's Rule is applied when we encounter an indeterminate form ∞/∞. By taking derivatives, we simplify the limit to a form that can be computed directly. In this example, as x approaches infinity, the limit of x / e^x becomes 0.

Squeeze Theorem

The Squeeze Theorem, also known as the Sandwich Theorem, is a fundamental concept in calculus used to determine the limit of a function by comparing it between two other functions. This theorem is particularly useful when dealing with functions that oscillate or fluctuate around a specific point. The Squeeze Theorem is used to establish the limit of a function within a certain range by "squeezing" it between two other functions whose limits are known. Mathematically, the Squeeze Theorem can be expressed as follows:

Suppose f(x), g(x), and h(x) are functions such that for all x in an interval containing 'c' (except possibly at 'c' itself), and f(x) ≤ g(x) ≤ h(x).

If

$$\lim_{x \to c} f(x) = \lim_{x \to c} h(x) = L$$

then:

$$\lim_{x \to c} g(x) = L$$

In other words, if the function g(x) is bounded between two other functions, f(x) and h(x), whose limits both converge to 'L', then the limit of g(x) as x approaches 'c' is also 'L'.

The Squeeze Theorem is a useful tool for finding the limit of functions that might not be directly evaluable using basic limit properties. It relies on the concept of function comparison to deduce the limit value of the function of interest.

Example 1:

Consider the limit: $\lim_{x \to 0} x^2 \, sin(1 \, / \, x)$

Using the Squeeze Theorem, we notice that

$$-x^2 \leq x^2 \, sin \frac{1}{x} \leq x^2$$

for all x ≠ 0

Taking the limit as x approaches 0 of these expressions:

$$\lim_{x\to 0}(-x^2) = 0$$

$$\lim_{x\to 0} x^2 = 0$$

Since $-x^2 \leq x^2 \sin\frac{1}{x} \leq x^2$ and both lower and upper bounds approach 0, by the Squeeze Theorem, we conclude:

$$\lim_{x\to 0} x^2 \sin\frac{1}{x} = 0$$

The Squeeze Theorem is used when we need to determine the limit of a function that's sandwiched between two other functions with known limits. In this example, as x approaches 0, the function $x^2 \sin\frac{1}{x}$ is bounded between $-x^2$ and x^2, both of which tend to 0, leading us to conclude that the limit of $x^2 \sin\frac{1}{x}$ is also 0.

Example 2:

Consider the limit: $\lim_{x\to 0} x^2 \cos\frac{1}{x} =$

Using the Squeeze Theorem, we notice that

$$-x^2 \leq x^2 \cos\frac{1}{x} \leq x^2$$

for all x ≠ 0

Taking the limit as x approaches 0 of these expressions:

$$\lim_{x\to 0}(-x^2) = 0$$

$$\lim_{x\to 0} x^2 = 0$$

Since $-x^2 \leq x^2 \cos\frac{1}{x} \leq x^2$ and both lower and upper bounds approach 0, by the Squeeze Theorem, we conclude:

$$\lim_{x\to 0} x^2 \cos\frac{1}{x} = 0$$

The Squeeze Theorem is applied to find the limit of a function that is bounded between two known functions. In this case, as x approaches 0, the function $x^2 \cos\frac{1}{x}$ is sandwiched between $-x^2$ and x^2, both of which tend to 0, leading us to conclude that the limit of $x^2 \cos\frac{1}{x}$ is also 0.

Examples

Example 1

Evaluate the limit as x approaches 2 for the function f(x) = 3x - 1.

Solution

To find the limit, we directly substitute the value of 'a' into the function:

Here a=2,

$$\lim_{x\to 2}(3x-1) = 3(2) - 1$$

= 6 - 1

= 5

Example 2

Evaluate the limit as x approaches 0 for the function g(x) = x^2.

Solution

a=0,

Substitute the value of 'a' into the function:

$$\lim_{x\to 0} x^2$$

= 0^2

= 0

Example 3

Evaluate the limit as x approaches 1 for the function $n(x) = \frac{x-1}{x^2-1}$

Solution

Factor the denominator and simplify:

$$h(x) = \frac{x-1}{(x+1)(x-1)}$$

$$= \lim_{x\to 1} \frac{x-1}{(x-1)(x+1)}$$

$$= \lim_{x\to 1} \frac{1}{x+1}$$

$$= 1 / 2$$

Example 4

Evaluate the limit as x approaches 3 for the function k(x) = $\left(\sqrt{x+1}\right) - 2$.

Solution

Substitute the value of 'a' into the function:

a=3,

$$\lim_{x\to 3} \sqrt{x+1} - 2$$

$$= \sqrt{3+1} - 2$$

$$= \sqrt{4} - 2$$

$$= 2 - 2$$

$$= 0$$

Example 5

Evaluate the limit as x approaches 2 for the function g(x) = $\frac{x^2-4}{x-2}$

Solution

Factor the numerator and simplify:

$$g(x) = \frac{(x+2)(x-2)}{(x-2)}$$

$$= \lim_{x \to 2} \frac{(x+2)(x-2)}{x-2}$$

$$= \lim_{x \to 2} (x+2)$$

$$= 2 + 2$$

$$= 4$$

Example 6

Evaluate the limit as x approaches 1 for the function h(x) = $\frac{x^3-1}{x^2-1}$

Solution

Factor both the numerator and denominator and simplify:

h(x) = $\frac{(x-1)(x^2-2x+1)}{(x-1)(x+1)}$

$$= \lim_{x \to 1} \frac{(x-1)(x^2-2x+1)}{(x-1)(x+1)}$$

$$= \lim_{x \to 1} x^2 + x + 1$$

$$= 1^2 + 1 + 1$$

$$= 3$$

Example 7

Evaluate the limit as x approaches 2 for the function f(x) = $\frac{x^2+2x-8}{x-2}$.

Solution

Factor the numerator and simplify:

$$f(x) = \frac{(x+4)(x-2)}{(x-2)}$$

$$= \lim_{x \to 2} \frac{(x+4)(x-2)}{(x-2)}$$

$$= \lim_{x \to 2} (x+4)$$

$$= 2 + 4$$

$$= 6$$

Example 8

Evaluate the limit as x approaches 0 for the function g(x) = (sin(x) / x).

Solution

Apply the Squeeze Theorem:

-1 ≤ sin(x) / x ≤ 1

As x approaches 0, both the upper and lower bounds approach 1, so the limit is 1.

Example 9

Evaluate the limit as x approaches 0 for the function k(x) = $\frac{e^x - 1}{x}$

Solution

Apply L'Hôpital's Rule:

$$= \lim_{x \to 0} \frac{e^x - 1}{x}$$

$$= \lim_{x \to 0} \frac{e^x}{1}$$

$$= 1 / 1$$

$$= 1$$

Example 10

Evaluate the limit as x approaches 0 for the function f(x) = (1 - cos(x)) / x.

Solution

Apply L'Hôpital's Rule:

$$\lim_{x \to 0} \frac{1 - \cos x}{x}$$

$$= \lim_{x \to 0} \frac{\sin(x)}{1}$$

$$= 0 / 1$$

$$= 0$$

5. CONTINUITY OF SUM OF TWO CONTINUOUS FUNCTIONS

The continuity of the sum of two continuous functions states that if two functions, f(x) and g(x), are individually continuous at a point 'a', then their sum,

$$h(x) = f(x) + g(x)$$

is also continuous at 'a'. In other words, when you add two continuous functions, the resulting function remains continuous.

Let's say we have two functions, f(x) and g(x), that are both continuous at a point 'a'. This means that as x approaches 'a', both f(x) and g(x) approach finite values, and there are no sudden jumps or breaks in their graphs.

When we consider the sum of these two functions,

$$h(x) = f(x) + g(x)$$

we observe that as x approaches 'a', both f(x) and g(x) are approaching finite values. Since addition is a basic arithmetic operation that preserves continuity, the sum of f(x) and g(x), represented by h(x), will also approach a finite value as x approaches 'a'. This ensures that there are no sudden discontinuities or jumps in the graph of h(x) at the point 'a'. Therefore,

h(x) = f(x) + g(x) is continuous at 'a'.

Example:

Suppose we have two continuous functions, f(x) = 2x + 3 and g(x) = x^2 - 1. Let's determine if their sum, h(x) = f(x) + g(x), is continuous at a point 'a' where a = 2.

Both f(x) = 2x + 3 and g(x) = x^2 - 1 are polynomial functions, and polynomial functions are continuous everywhere. Therefore, both functions are individually continuous at any point 'a', including a = 2.

Now, let's consider their sum,

$$h(x) = f(x) + g(x): h(x)$$

$$= (2x + 3) + (x^2 - 1)$$

$$= x^2 + 2x + 2$$

Since both f(x) and g(x) are continuous at a = 2, their sum,

h(x) = x^2 + 2x + 2 is also continuous at a = 2.

This means that as x approaches 2, the value of h(x) approaches

$$2^2 + 2 * 2 + 2 = 10$$

There are no sudden jumps or breaks in the graph of h(x) at x = 2.

6. CONTINUITY OF PRODUCT OF TWO CONTINUOUS FUNCTIONS

The continuity of the product of two continuous functions states that if two functions, f(x) and g(x), are individually continuous at a point 'a', then their product,

$$h(x) = f(x) * g(x)$$

is also continuous at 'a'. In other words, when you multiply two continuous functions, the resulting function remains continuous.

Explanation: Assuming that both f(x) and g(x) are continuous at a point 'a', this implies that as x approaches 'a', both functions approach finite values and there are no sudden breaks or gaps in their graphs.

Considering the product of these two functions,

$$h(x) = f(x) * g(x),$$

we can examine the behaviour as x approaches 'a'. Since multiplication is another basic arithmetic operation that preserves continuity, the product of f(x) and g(x), represented by h(x), will also approach a finite value as x approaches 'a'. This means that there are no abrupt changes or disruptions in the graph of h(x) at the point 'a'. Consequently,

$$h(x) = f(x) * g(x) \text{ is continuous at 'a'.}$$

Example:

Consider two continuous functions, f(x) = sin(x) and g(x) = e^x, where 'x' is in the interval [0, π/2]. Let's investigate the continuity of their product,

$$h(x) = f(x) * g(x).$$

Both f(x) = sin(x) and g(x) = e^x are continuous functions in their respective domains. The sine function and the exponential function are known to be continuous everywhere. Since the interval [0, π/2] lies within the domain of both functions, they are continuous in this interval.

Now, let's examine their product, h(x) = f(x) * g(x): h(x) = sin(x) * e^x

Because both f(x) and g(x) are continuous in the interval [0, π/2], their product h(x) = sin(x) * e^x is also continuous in this interval. This implies that as x approaches any value within [0,

π/2], the product h(x) = sin(x) * e^x approaches a finite value, and there are no abrupt changes or disruptions in the graph of h(x) within this interval. Thus, the product of two continuous functions remains continuous within the interval [0, π/2].

7. CONTINUITY OF DIFFERENCE OF TWO CONTINUOUS FUNCTIONS

The continuity of the difference of two continuous functions states that if two functions, f(x) and g(x), are individually continuous at a point 'a', then their difference,

$$h(x) = f(x) - g(x)$$

is also continuous at 'a'. In other words, when you subtract two continuous functions, the resulting function remains continuous.

Explanation: If both f(x) and g(x) are continuous at a particular point 'a', it means that as x approaches 'a', both functions approach finite values, and there are no sudden jumps or breaks in their graphs.

When we consider the difference of these two functions, h(x) = f(x) - g(x), we analyse how the function behaves as x approaches 'a'. Subtraction is a fundamental arithmetic operation that maintains continuity, so the difference of f(x) and g(x), represented by h(x), will also approach a finite value as x approaches 'a'. This indicates that there are no abrupt discontinuities or gaps in the graph of h(x) at the point 'a'.

Therefore,

h(x) = f(x) - g(x) is continuous at 'a'.

Consider two continuous functions,

f(x) = $3x^2$ - 2x and g(x) = x^2 + 5x.

Let's analyse the continuity of their difference,

h(x) = f(x) - g(x), at the point 'a' where a = 2.

Both f(x) = 3x^2 - 2x and g(x) = x^2 + 5x are polynomial functions, which are known to be continuous everywhere. This means that they are individually continuous at any point 'a', including a = 2.

Now, let's examine their difference,

$$h(x) = f(x) - g(x): h(x) = (3x^2 - 2x) - (x^2 + 5x) = 2x^2 - 7x$$

Since both f(x) and g(x) are continuous at a = 2, their difference

$$h(x) = 2x^2 - 7x$$

is also continuous at a = 2. As x approaches 2, the value of h(x) approaches

$$2 * 2^2 - 7 * 2 = -6.$$

There are no sudden disruptions or gaps in the graph of h(x) around x = 2.

8. CONTINUITY OF QUOTIENT OF TWO CONTINUOUS FUNCTIONS

The continuity of the quotient of two continuous functions states that if two functions, f(x) and g(x), are individually continuous at a point 'a', and g(a) ≠ 0, then their quotient,

$$h(x) = f(x) / g(x)$$

is also continuous at 'a'. In other words, when you divide two continuous functions, the resulting function remains continuous if the denominator is not zero at 'a'.

Explanation: Assuming both f(x) and g(x) are continuous at a specific point 'a', and further assuming that g(a) ≠ 0 (denominator is not zero at 'a'), this implies that as x approaches 'a', both functions approach finite values, and there are no sudden gaps or breaks in their graphs.

When we consider the quotient of these two functions,

$$h(x) = f(x) / g(x)$$

we examine how the function behaves as x approaches 'a'. Division is another basic arithmetic operation that maintains continuity, but there's an additional requirement: the denominator must not be zero at 'a'. As long as g(a) ≠ 0, the quotient of f(x) and g(x), represented by h(x), will approach a finite value as x approaches 'a'. This guarantees that there are no abrupt discontinuities or gaps in the graph of,

h(x) = f(x) / g(x) is continuous at 'a' when g(a) ≠ 0.

Example:

Let's examine the continuity of the quotient of two continuous functions. Consider

$$f(x) = 1/x \text{ and } g(x) = x - 2.$$

We will investigate the continuity of their quotient,

$$h(x) = f(x) / g(x),$$

for values of 'x' where the denominator $g(x) \neq 0$.

The function $f(x) = 1/x$ is continuous for all $x \neq 0$, and the function $g(x) = x - 2$ is continuous for all real values of x. We need to ensure that the denominator $g(x) \neq 0$, which means $x \neq 2$ to avoid division by zero.

Now, let's analyse their quotient,

$$h(x) = f(x) / g(x): h(x) = (1/x) / (x - 2) = 1 / (x(x - 2))$$

Since both f(x) and g(x) are continuous where they are defined and the denominator $x(x - 2)$ is nonzero in the interval $(-\infty, 0) \cup (0, 2) \cup (2, \infty)$, the quotient $h(x) = 1 / (x(x - 2))$ is also continuous in these intervals. This means that as x approaches any value within the specified intervals, the value of h(x) remains well-defined and continuous. However, h(x) has a vertical asymptote at $x = 0$ and a removable discontinuity at $x = 2$, where the denominator becomes zero.

9. SUMMARY

In this unit, we kicked things off with composite functions, this fascinating concept where we take the output of one function and feed it as the input to another. It's like a relay race in mathematics, where the baton is passed from one function to the next, creating a chain of operations that result in a new, composite function. This concept is not just mathematically elegant but also mirrors processes in computer programming, where functions often work in tandem, passing results along to build up complex operations.

Then, we danced through the operations on functions – sum, difference, product, and quotient. Each operation added a new layer to our understanding, much like adding ingredients to a recipe changes the flavour of a dish. These operations allow us to combine functions in various ways, crafting new functions that can describe a wide range of phenomena, from the trajectories of planets to the fluctuations in stock markets.

Our foray into periodic functions introduced us to the rhythms of mathematics, where certain functions repeat their values in regular intervals, much like the cycles of seasons or the ticking of a clock. This concept helps us grasp phenomena that recur over time, providing a framework for predicting future behaviour based on past patterns.

Diving into the world of even and odd functions added another dimension to our understanding, showing us how functions can mirror or rotate around axes, revealing symmetrical patterns akin to reflections in water or shadows cast by the sun.

As we transitioned into the concepts of limits and continuity, we ventured closer to the heart of calculus. Limits allowed us to probe the behaviour of functions as they approach specific points, offering insights into how functions behave under constrained conditions. This is crucial for understanding how algorithms perform, especially as they near boundary conditions.

Continuity ensured that our functions behaved nicely, without any unexpected jumps or breaks. This seamless behaviour is essential in programming, where we expect algorithms to execute smoothly, without any surprises.

In wrapping up this unit, it's clear that the journey through composite functions, operations, limits, and continuity isn't just a mathematical excursion. It's a foundational toolkit for anyone delving into the world of computer applications. These concepts equip us with the ability to predict, model, and analyse the intricate behaviours of algorithms and systems, ensuring we can navigate the complexities of the digital world with confidence and precision.

10. SELF-ASSESSMENT QUESTIONS

SELF-ASSESSMENT QUESTIONS – 1

1. What does the limit of a function represent as the input approaches a particular point?
   a) The value of the function at that point
   b) The maximum value of the function
   c) The value the function approaches as the input gets closer to the point
   d) The minimum value of the function
2. In the context of calculus, what does the concept of limit characterize?
   a) The absolute maximum value of a function
   b) The behavior of a function as its input values converge towards a point
   c) The point where a function has its minimum value
   d) The average value of a function over an interval
3. Which mathematical notation is used to represent the limit of a function as it approaches a specific point 'a'?
   a) $\lim(x \to a)\ f(x)$
   b) $f(a) = \lim(x \to 0)\ f(x)$
   c) $\lim(f(x)) = a$
   d) $f(a) = x \to a$

4. When is a function said to be continuous at a point 'a'?
   a) When the function has a sharp corner at 'a'
   b) When the function's value at 'a' is undefined
   c) When the limit of the function at 'a' does not exist
   d) When the limit of the function at 'a' exists and equals the value of the function at 'a'
5. What does continuity of a function guarantee about its behaviour at a specific point?
   a) There are abrupt changes in the graph of the function
   b) The function's values experience sudden alterations
   c) The function's values transition smoothly and without breaks
   d) The function's values diverge as the input approaches the point
6. What does the ε-δ definition of limits provide a formal way to express?
   a) The concept of differentiation
   b) The behaviour of a function at infinity
   c) The concept of limit in calculus
   d) The behaviour of a function at critical points
7. According to the ε-δ definition of limits, for a given function f(x) and point 'a', what is the condition for the limit to be 'L' as x approaches 'a'?
   a) For every positive ε, there exists a positive δ such that $|x - a| < \delta$ implies $|f(x) - L| < \varepsilon$.
   b) For every positive δ, there exists a positive ε such that $|x - a| < \delta$ implies $|f(x) - L| < \varepsilon$.
   c) For every positive ε, there exists a positive δ such that $|f(x) - L| < \varepsilon$ implies $|x - a| < \delta$.
   d) For every positive δ, there exists a positive ε such that $|f(x) - L| < \delta$ implies $|x - a| < \varepsilon$.

8. What does the Sum Rule state in the context of limits of functions?
   a) The limit of the product of two functions is the product of their individual limits.
   b) The limit of the difference between two functions is the difference of their individual limits.
   c) The limit of the sum of two functions is the sum of their individual limits.
   d) The limit of the quotient of two functions is the quotient of their individual limits.
9. According to the Product Rule, what is the limit of the product of two functions f(x) and g(x) as x approaches 'a'?
   a) The product of their individual limits at 'a'.
   b) The sum of their individual limits at 'a'.
   c) The difference of their individual limits at 'a'.
   d) The quotient of their individual limits at 'a'.
10. What does the Quotient Rule for limits state?
   a) The limit of the product of two functions is the product of their individual limits.
   b) The limit of the sum of two functions is the sum of their individual limits.
   c) The limit of the quotient of two functions is the quotient of their individual limits, regardless of the denominator.
   d) The limit of the quotient of two functions is the quotient of their individual limits, provided the denominator's limit is not zero.
11. What is the primary purpose of L'Hôpital's Rule in calculus?
   a) To find the derivative of a given function.
   b) To simplify limits involving indeterminate forms using derivatives.
   c) To calculate the integral of a function.
   d) To determine the continuity of a function.

12. When can L'Hôpital's Rule be applied to evaluate a limit?
    a) Whenever the limit involves fractions.
    b) When both the numerator and denominator approach zero or infinity, and the derivatives of both exist.
    c) Only for limits at positive infinity.
    d) Only for limits at negative infinity.
13. What does the Squeeze Theorem, or the Sandwich Theorem, help determine in calculus?
    a) The value of a definite integral.
    b) The limit of a function by comparing it between two other functions.
    c) The derivative of a function.
    d) The local maximum and minimum of a function.
14. What does the continuity of the sum of two continuous functions state?
    a) The sum of two continuous functions is always discontinuous.
    b) If two functions are individually continuous at a point, their sum is also continuous at that point.
    c) If two functions are individually discontinuous, their sum is continuous at that point.
    d) The sum of two continuous functions is only continuous at the origin.
15. Why is the sum of two continuous functions continuous at a point 'a'?
    a) Because addition is a basic arithmetic operation.
    b) Because both functions have a limit at 'a'.
    c) Because it ensures there are sudden jumps in the graph.
    d) Because it only works for specific values of 'a'.

16. What does the continuity of the product of two continuous functions state?
    a) The product of two continuous functions is always discontinuous.
    b) If two functions are individually continuous at a point, their product is also continuous at that point.
    c) If two functions are individually discontinuous, their product is continuous at that point.
    d) The product of two continuous functions is only continuous at the origin.
17. Why is the product of two continuous functions continuous at a point 'a'?
    a) Because multiplication is a basic arithmetic operation.
    b) Because both functions have a limit at 'a'.
    c) Because it ensures there are sudden breaks in the graph.
    d) Because it only works for specific values of 'a'.
18. What does the continuity of the difference of two continuous functions state?
    a) The difference of two continuous functions is always discontinuous.
    b) If two functions are individually continuous at a point, their difference is also continuous at that point.
    c) If two functions are individually discontinuous, their difference is continuous at that point.
    d) The difference of two continuous functions is only continuous at the origin.
19. Why is the difference of two continuous functions continuous at a point 'a'?
    a) Because subtraction is a basic arithmetic operation.
    b) Because both functions have a limit at 'a'.
    c) Because it ensures there are sudden breaks in the graph.
    d) Because it only works for specific values of 'a'.
20. What condition must be satisfied for the quotient of two continuous functions to be continuous at a point 'a'?
    a) Both functions must be discontinuous at 'a'.
    b) The numerator must be zero at 'a'.
    c) The denominator must be zero at 'a'.

11. TERMINAL QUESTIONS

1. Define what it means for a function to be continuous at a specific point 'a'. Provide an example to illustrate continuity.
2. Explain the significance of continuity in mathematics and provide an example of how it's used in a real-world application.
3. State the First Principle of Limits (ε-δ definition) and explain its significance in understanding limits.
4. Discuss the properties of continuous functions, including the continuity of sums, products, differences, and quotients of continuous functions. Provide examples to illustrate each property.
5. Given the functions f(x) = 2x + 1 and g(x) = 3x^2 - 4, determine whether their sum, product, difference, and quotient (where defined) are continuous functions. Provide explanations.
6. Apply the Squeeze Theorem to find the limit as x approaches 0 of the function g(x) = x^2 * sin(1/x). Show all necessary steps and justify your answer.
7. Solve the limit as x approaches 2 of the function f(x) = (x^2 - 4) / (x - 2). Justify your answer.
8. Determine whether the function f(x) = sqrt(x) is continuous at x = 4. Explain your reasoning.
9. Use the properties of continuity to determine whether the function h(x) = e^x * sin(x) is continuous for all real numbers. Provide a proof.
10. Prove or disprove the following statement: "The difference of two continuous functions is always continuous."

12. ANSWERS

Self-assessment answers

1. c) The value the function approaches as the input gets closer to the point
2. b) The behaviour of a function as its input values converge towards a point
3. a) $\lim(x \to a) f(x)$
4. d) When the limit of the function at 'a' exists and equals the value of the function at 'a'
5. c) The function's values transition smoothly and without breaks
6. c) The concept of limit in calculus
7. a) For every positive ε, there exists a positive δ such that $|x - a| < \delta$ implies $|f(x) - L| < \varepsilon$.
8. c) The limit of the sum of two functions is the sum of their individual limits.
9. a) The product of their individual limits at 'a'.
10. d) The limit of the quotient of two functions is the quotient of their individual limits, provided the denominator's limit is not zero.
11. b) To simplify limits involving indeterminate forms using derivatives.
12. b) When both the numerator and denominator approach zero or infinity, and the derivatives of both exist.
13. b) The limit of a function by comparing it between two other functions.
14. b) If two functions are individually continuous at a point, their sum is also continuous at that point.
15. a) Because addition is a basic arithmetic operation that preserves continuity.
16. b) If two functions are individually continuous at a point, their product is also continuous at that point.
17. a) Because multiplication is a basic arithmetic operation that preserves continuity.
18. b) If two functions are individually continuous at a point, their difference is also continuous at that point.
19. a) Because subtraction is a fundamental arithmetic operation that maintains continuity.
20. d) The denominator must not be zero at 'a'.

Terminal Answers

1. A function f(x) is continuous at a point 'a' if the following three conditions are met:

   f(a) is defined (i.e., the function has a value at 'a').

The limit of the function as x approaches 'a' exists.

The limit as x approaches 'a' is equal to the value of the function at 'a': lim(x→a) f(x) = f(a).

Example: Consider the function f(x) = 2x + 3. This function is continuous at any point 'a' because it satisfies all three conditions. For instance, it is continuous at 'a = 1' because f(1) = 5, $\lim_{x\to 1} f(x)$ = 5, and $\lim_{x\to 1} f(x)$ = f(1).

2. Continuity is a crucial concept in mathematics because it ensures that functions behave predictably and smoothly. In real-world applications, continuity is essential in areas like physics, engineering, and economics. For instance, in physics, the continuous motion of an object can be described by continuous functions. In engineering, continuous functions are used to model behaviours such as stress in materials. In economics, continuous functions are used to model demand and supply curves, ensuring that transitions are smooth and not abrupt.
3. The First Principle of Limits states that for a function f(x), the limit as x approaches a point 'a' is 'L' if, for any positive ε (epsilon), there exists a positive δ (delta) such that if the distance between x and a (|x - a|) is less than δ, then the distance between f(x) and L (|f(x) - L|) is less than ε. This principle is significant because it provides a rigorous and precise definition of limits, allowing us to understand how a function behaves as it approaches a specific point.
4. **Sum Rule**: The sum of two continuous functions is continuous. For example, if f(x) = x^2 and g(x) = 2x, both are continuous, and their sum h(x) = x^2 + 2x is also continuous.

   Product Rule: The product of two continuous functions is continuous. If f(x) = sin(x) and g(x) = cos(x), both are continuous, and their product h(x) = sin(x) * cos(x) is continuous.

   Difference Rule: The difference of two continuous functions is continuous. For example, if f(x) = 3x and g(x) = x^2, both are continuous, and their difference h(x) = 3x - x^2 is continuous.

   Quotient Rule: The quotient of two continuous functions is continuous where the denominator is not zero. If f(x) = 1/x and g(x) = x, both are continuous, and their quotient h(x) = (1/x) / x = $1/x^2$ is continuous where x ≠ 0.

5. **Sum**: The sum h(x) = f(x) + g(x) = (2x + 1) + ($3x^2$ - 4) is continuous because it's the sum of two continuous functions.

   Product: The product h(x) = f(x) * g(x) = (2x + 1)($3x^2$ - 4) is continuous because it's the product of two continuous functions.

   Difference: The difference h(x) = f(x) - g(x) = (2x + 1) - ($3x^2$ - 4) is continuous because it's the difference of two continuous functions.

   Quotient: The quotient h(x) = f(x) / g(x) = (2x + 1) / ($3x^2$ - 4) is continuous where g(x) ≠ 0. To check where it's defined, set the denominator equal to zero and solve: $3x^2$ - 4 = 0 → x = ±2/√3. Therefore, h(x) is continuous except at x = 2/√3 and x = -2/√3.

6. To apply the Squeeze Theorem, we need to find two functions, f(x) and h(x), such that:

   f(x) ≤ g(x) ≤ h(x) for all x in a neighbourhood of 0.

   The limits of f(x) and h(x) as x approaches 0 are equal and finite.

   Let's choose f(x) = $-x^2$and h(x) = x^2.

   For f(x) ≤ g(x) ≤ h(x):

   $-x^2$≤ x^2* sin(1/x) ≤ $x^2$for all x ≠ 0.

   Now, let's find the limits of f(x) and h(x):

$$\lim_{x\to 0} f(x) = \lim_{x\to 0} -x^2 = 0$$

$$\lim_{x\to 0} h(x) = \lim_{x\to 0} x^2 = 0$$

   Since both limits of f(x) and h(x) are equal and finite, by the Squeeze Theorem, $\lim_{x\to 0} g(x) = 0$

7. To solve this limit, we can attempt direct substitution. However, if we substitute x = 2 directly, we get 0/0, which is an indeterminate form. Therefore, we can apply L'Hôpital's Rule.

   Find the derivatives of the numerator and denominator:

   f'(x) = d/dx (x^2- 4) = 2x

   g'(x) = d/dx (x - 2) = 1

   Apply L'Hôpital's Rule:

$$\lim_{x\to 2} \frac{x^2 - 4}{x - 2} = \lim_{x\to 2} \frac{f'(x)}{g'(x)} = \lim_{x\to 2} \frac{2x}{1}$$

   Now, we can directly substitute x = 2:

$$\lim_{x\to 2}\frac{2x}{1} = \frac{2(2)}{1} = 4$$

So, $\lim_{x\to 2}\frac{x^2-4}{x-2} = 4$

8. To determine if the function f(x) = sqrt(x) is continuous at x = 4, we need to check the three conditions for continuity:

   f(4) is defined: f(4) = $\sqrt{4}$= 2, so f(4) is defined.

   The limit of f(x) as x approaches 4 exists: $\lim_{x\to 4}\sqrt{x}$also equals 2.

   f(4) equals the limit: f(4) = 2, which matches the limit.

   Since all three conditions are met, f(x) = $\sqrt{x}$is continuous at x = 4.

9. To determine the continuity of h(x) = e^x* sin(x) for all real numbers, we need to show that it's continuous everywhere. $e^x$is a continuous function for all real numbers.

   sin(x) is also a continuous function for all real numbers.

   Since both e^x and sin(x) are continuous everywhere, the product of two continuous functions, h(x) = e^x * sin(x), is continuous for all real numbers. This is because the properties of continuity state that the product of continuous functions is continuous.

10. The statement is true. Let's prove it:

    Suppose we have two continuous functions, f(x) and g(x). We want to show that their difference, h(x) = f(x) - g(x), is also continuous. To prove this, we can use the properties of continuous functions:

    Both f(x) and g(x) are continuous by definition.

    The sum of two continuous functions is continuous (as shown previously).

    Multiplication of a continuous function by a constant is continuous.

    Now, we can express h(x) as a sum: h(x) = f(x) + (-1) * g(x).

    Since addition and multiplication by a constant preserve continuity, both f(x) and -g(x) are continuous. Therefore, their sum, h(x), is continuous.

    So, the difference of two continuous functions is indeed always continuous.